

KS

PRODUCTION-READY ARCHITECTURE · V1.4.0

Karobar Saathi (■■■■■■■■■■■■■■■■■■■■)

Spoken Transactions -> Confirmed Ledger -> Explainable Financial Evidence

Bridging Pakistan's \$180B informal economy with AI voice bookkeeping and sovereign credit profiles for 5 Million+ micro-merchants.

STACK: Flutter (Riverpod) · FastAPI · Groq Whisper-large-v3 · Vernacular LLM Parser · SQLite

STATUS: Live Backend & APK Release v1.4.0 | **TARGET:** Chai Stalls, Kirana Stores, Home Tailors, Khokhas

LICENSE: MIT Open Source | **WEBSITE:** ahsankhizar5.github.io/karobar-saathi

The Invisible 5 Million & The Unbanked Trap

Micro-businesses form the economic spine of Pakistan, yet 90%+ operate completely invisible to formal finance.

5.2M+

Informal Micro-Enterprises

Khokhas, dhabas, roadside carts, and home workers handling daily cash volume without bank accounts or formal registrations.

93%

Paper Only (Bahi Khata)

Traditional accounting apps fail. Typing in English or Urdu after 14 hours of shop work is slow, confusing, and exhausting.

25-30%

Monthly Loan Shark Rates

Without verified financial history, commercial banks reject micro-loans. Shopkeepers are trapped relying on informal moneylenders.

THE CORE BOTTLENECK:

Banks don't lend without evidence. Shopkeepers can't generate evidence without unbearable bookkeeping friction.

Zero Typing. Zero Complexity. Just Speak in Urdu.

Karobar Saathi replaces the tedious bahi khata with an AI assistant that understands Pakistani bazaar vernacular.

1. Natural Voice-to-Ledger (5 Seconds a Day)

Speak at closing time. Groq Whisper-v3 + LLM parse complex multi-item sales and expenses in one breath with zero typing.

2. Hallucination-Proof Roman Urdu Clarification

Never invents numbers. If ambiguous (e.g. '3000 diye'), prompts the shopkeeper: 'Bikri ki amount kya hai ya kisko diye?' before writing to disk.

3. Sovereign, Consent-Gated Evidence Engine

Transforms 30 days of daily entries into underwriting evidence (sales volatility, cash buffer runway) for microfinance banks — with instant one-tap revocation.

The 5-Second Shopkeeper Flow

■ Tap Mic

Shopkeeper records voice note in Urdu/Roman Urdu.

■ 1.2s Transcription

Hosted Whisper-large-v3 converts voice to text.

■ Vernacular Parser

Pakistani bazaar terms extracted into strict JSON schema.

■ One-Tap Confirm

Visual sheet shows sale/purchase before writing to ledger.

■ Instant Signals

Net profit, 7-day sales trend, and loan readiness update.

Engineered for Low-End Android & Slow Connections

A resilient three-tier pipeline connecting shopkeeper smartphones to institutional lending APIs.

LAYER 1: PRESENTATION & RECORDING (FLUTTER APP)

State-managed with Riverpod. WhatsApp-style live waveform recording (record: ^5.2.1), haptic feedback, bilingual EN/UR with full RTL layout, and 150s cold-start tolerance.

Tech: Flutter 3.x · Riverpod · record ^5.2.1 · Urdu RTL Layout

LAYER 2: VERNACULAR AI PIPELINE (FASTAPI ENGINE)

Groq Whisper-large-v3 speech-to-text combined with specialized Pakistani bazaar LLM parser. Enforces strict JSON Schema and flags ambiguous inputs for conversational clarification.

Tech: FastAPI · Groq Whisper-v3 · Groq LLM · Bazaar Vernacular Glossary

LAYER 3: LEDGER STORE & EVIDENCE COMPUTER (B2B API)

ACID SQLite persistence storing confirmed entries and original spoken audio transcripts. Computes 30-day volatility (CV), cash buffer runway, and loan ranges. Protected by X-User-Consent.

Tech: SQLite · Rule Heuristic Engine · X-User-Consent · REST API

Deciphering Pakistani Bazaar Vernacular

Standard financial LLMs fail in street Urdu. Karobar Saathi is calibrated with a native bazaar taxonomy.

Sales & Income -> type: "sale"

"Bikri", "Kamai", "Kamaya", "Aamdani", "Becha", "Ki sale hui"

Stock Purchases -> type: "purchase"

"Maal", "Stock liya", "Khareeda", "Samaan", "Cheeni doodh liya"

Expenses & Bills -> type: "expense"

"Bijli ka bill", "Dukaan ka kiraya", "Gas bill", "Committee"

Withdrawals -> type: "withdrawal"

"Ghar bheje", "Ghar ke liye nikaale", "Ghar kharch", "Udhaar diya"

Multi-Transaction Extraction in One Sentence

A shopkeeper speaks in a single breath:

"Aaj 6000 ki bikri hui, 2500 ka stock liya, aur 500 kiraya diya."

Deconstructed into 3 Distinct Accounting Entries:

■ Sale: PKR 6,000

Category: other | Note: Aaj bikri hui

■ Purchase: PKR 2,500

Category: stock | Note: Stock liya

■ Expense: PKR 500

Category: rent | Note: Kiraya diya

The Zero-Hallucination Safeguard

In financial bookkeeping, AI hallucinations corrupt balance sheets. Karobar Saathi never guesses.

The Ambiguous Input: "3000 diye"

A shopkeeper says: '3000 diye' (Gave 3000).
Did they buy inventory? Pay rent? Give cash to family?

■ Generic AI:

Guesses 'Expense: PKR 3,000' (Silently corrupts the ledger).

✓■ Karobar Saathi:

Flags entry_type: 'unclear' & disables saving until clarified.

Conversational Clarification Prompt:

"Bikri ki amount kya hai ya ye paise kisko diye the?"

Three Principles of Financial Integrity

1. Strict Structural Output Enums

JSON Schema enforcement: entry_type must strictly be one of [sale, purchase, expense, withdrawal, unclear].

2. Human-in-the-Loop Confirmation

Every extracted transaction renders an editable visual card. The user can tweak category or amount before saving.

3. Spoken Audio Audit Trail

Original spoken audio transcripts are permanently bound to the ledger entry, creating an immutable forensic record.

Converting Voice Notes into Underwriting Evidence

Microfinance banks don't have time to read thousands of ledger lines. We distill raw records into statistical signals.

1. Transaction Consistency Index

Calculates recorded active days over 30 days. Proves operational stability.

Formula: $\text{Active Days} / 30$ (Target: > 80% for Tier 1 micro-loan)

2. Sales Volatility (Coefficient of Variation)

Evaluates week-over-week revenue variance to assess income stability.

Formula: $\text{StdDev}(\text{Sales}) / \text{Mean}$ ($\text{CV} < 0.20$ = Low Risk)

3. Net Cash Buffer Runway

Measures how many days of operating expenses are covered.

Formula: $\text{Net Cash} / \text{Daily Expenses} = \text{Buffer Days}$ (Safe: > 7 days)

Lender Evidence Profile API Output

GET /api/v1/evidence-profile/shop_001

```
{
  "user_id": "shop_001",
  "avg_daily_sales": 4250.00,
  "sales_volatility": "low",
  "days_with_transactions": 26,
  "cash_buffer_days": 8,
  "net_cash_position": 48200.00,
  "readiness_summary": "Stable cash flow.
    Suitable for micro-loan PKR 50,000-100,000.",
  "explainable_factors": [
    "Recorded transactions on 87% of days in last 30 days",
    "Sales variance < 20% week-to-week – stable pattern",
    "Positive cash buffer: ~8 days expenses covered"
  ]
}
```

Shopkeeper Sovereignty: Consent-Gated Architecture

We refuse predatory data harvesting. The shopkeeper retains absolute legal and technical ownership of their financial records.



Cryptographic Consent

Evidence API strictly requires an X-User-Consent: true header verified against database user state.



One-Tap Revocation

If the shopkeeper toggles consent off, external API queries are immediately blocked with 403.



No Ad Tracking

No advertising SDKs, no behavioral trackers. Data is never shared or sold to external brokers.

LIVE PROTOCOL VERIFICATION ([docs/VERIFICATION.md](#)):

Revoking consent via PATCH .../consent makes GET /api/v1/evidence-profile return 403 consent_required even with X-User-Consent: true.

TARGET MARKET PERSONAS

Designed for Pakistan's Real Retail Ecosystem

Calibrated with 30-day realistic test datasets representing the three largest segments of informal commerce.

■ Ahmad Chai Wala

Tea Stall · Urban Bazaar

Daily Sales: PKR 3,000 - 5,500

Consistency: 85% (High velocity)

Top Costs: Milk, sugar, tea, LPG

"Needs quick 25k working capital for tea stock during winter"

■ Bibi Naseem

Kirana Store · Neighborhood

Daily Sales: PKR 5,000 - 9,000

Consistency: 93% (Very consistent)

Top Costs: Wholesale dry goods & FMCG

"Needs 100k credit line to buy wholesale dry goods at bulk d"

■ Fatima Silai

Home Tailor · Female Solo

Daily Sales: PKR 1,500 - 4,000

Consistency: 72% (Seasonal spikes)

Top Costs: Fabric, thread, machine repair

"Home-based woman entrepreneur unable to travel to bank branch"

Production-Grade Rigor & Battle-Tested Code

Not just a prototype — backed by comprehensive test coverage, cold-start handling, and verified audio pipelines.

■ Render Cold-Start Resilience

Extended LLM call timeouts from 30s to 150s with localized status messages.
Background keep-warm pings every 10 mins eliminate user-facing latency.

■ Hardware Voice-Recorder Migration (v1.4.0)

Migrated to the umbrella record: ^5.2.1 package for 100% Android hardware reliability. Added optimistic state, waveform animation, and 60s auto-stop.

■ Automated GitHub Actions CI

Continuous integration runs full backend pytest suites alongside Flutter analyzer and widget tests on every pull request.

Verification Scorecard

100%

Flutter Analyze Clean

11 / 11

Widget & Unit Tests Passing

6 / 6

Seed Audio Files E2E Verified

v1.4.0

Release APK Ready for Install

High-Margin B2B FinTech Business Model

100% free for micro-businesses. Monetization is driven by institutional lenders and wholesale distributors.

SHOPKEEPER LAYER

Always Free App

Zero subscription fees for shopkeepers. Removes all friction to achieve viral word-of-mouth adoption across bazaar trade associations.

LENDER B2B API

Credit Origination Fees

Microfinance Banks (Mobilink Bank, Akhuwat, Kashf) pay a c
ry fee per Evidence Profile plus a 1.5% - 2.5% origination f
ee upon loan disbursement.

SUPPLY FINANCING

FMCG Bulk Credit

FMCG distributors (Unilever, Nestlé, Engro) utilize stock pu
rchase evidence to underwrite 7-day inventory credit directl
y inside the app.

GO-TO-MARKET FLYWHEEL:

Wholesale Mandi Onboarding -> Voice Recording Daily Habit -> First 30-Day Milestone -> Instant Loan Qualification | \$180B+ Market

Strategic Product & Expansion Roadmap

Scaling from working proof-of-concept to national financial inclusion infrastructure.

PHASE 1 (Current · Q3 2026)

Offline-First Voice Sync

- Local encrypted SQLite audio queue for zero-connectivity.
- Background auto-sync when cellular signal returns.
- Resilient auto-scaling cloud cluster deployment.

PHASE 2 (Q4 2026)

Multi-Dialect Expansion

- Fine-tuned Whisper models for Pashto, Sindhi, Punjabi.
- Voice-based WhatsApp bot integration.
- Automated customer ledger reminders (Udhaar recovery).

PHASE 3 (Q1 2027)

Raast Payment Rails

- State Bank of Pakistan Raast P2M QR rails integration.
- One-click digital loan acceptance & micro-repayments.
- Formal credit bureau data ingestion (1Link / SBP ECIB).

THE VISION

Dukaan Aapki, Hisaab Karobar Saathi Ka.



Empowering 5 million hardworking micro-entrepreneurs to be seen, verified, and funded — using nothing more than their daily voice.

■ Explore the Project & Resources:

- **Download APK v1.4.0:** github.com/ahsankhizar5/karobar-saathi/releases/latest
- **Live API Homepage:** ahsankhizar5.github.io/karobar-saathi
- **Production Backend:** karobar-saathi.onrender.com